

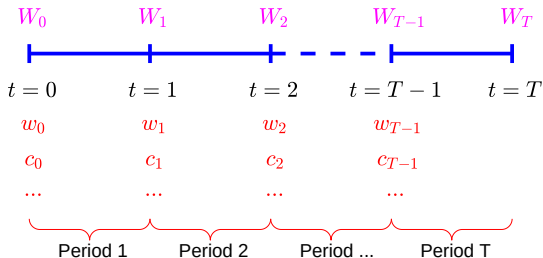
QUANTITATIVE FINANCE

PART III - DYNAMIC DECISIONS

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The Multi-Period Setup



At the current time $t = 0$, given the initial wealth W_0 , we decide the optimal investment (w_t , $t = 0, 1, \dots, T - 1$), consumption (c_t , $t = 0, 1, \dots, T - 1$), and possibly other policies (e.g., life insurance, working hours) for each period at the beginning of that period. These decisions in turn determine the wealth accumulation (W_t , $t = 1, 2, \dots, T$).

The policy variables are called *control variables* and the variables that measure the “state” of the dynamic environment of decisions, that is, provide information about the system’s history to enable computation of its future behavior, are called *state variables*. For example, the wealth level W_t above is a state variable.

Consider the following consumption-savings problem.

$$\begin{aligned}
 \max_{\{c_t\}_{t=0}^{T-1}} \quad & \sum_{t=0}^{T-1} \sqrt{c_t} + \sqrt{W_T}, \\
 \text{s.t.} \quad & W_{t+1} = (W_t - c_t)(1 + r), \quad t = 0, 1, \dots, T-1,
 \end{aligned} \tag{1}$$

where r is the constant savings rate.

The decision c_0 will impact not only the reward/felicity ($\sqrt{c_0}$) for the present period 1, but also those for all the future periods through wealth accumulation.¹ This is the reason that the problem is truly a dynamic one.

Generally, control variables control the change of state variables. The special structure between control and state variables at successive time points enables us to say interesting things about the solutions.

¹The last term $\sqrt{W_T}$ warrants a different interpretation, usually referred to as bequest, which does not necessarily take the same functional form as that of the flow consumption.

Dynamic Optimization

There are three major approaches to tackle dynamic optimization problems.

- 1 Calculus of Variations
- 2 Optimal Control
- 3 Dynamic Programming

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The Euler Equation

Its solution renders the famous **Euler equation** which is omnipresent in dynamic problems in financial economics.

The Euler Equation

The optimal state path satisfies the necessary condition:

$$F_x - \frac{dF_{x'}}{dt} = 0, \quad (3)$$

where the first term is a partial derivative and the second term is a total derivative (i.e., $\frac{dF_{x'}}{dt} = F_t + F_x x' + F_{x'} x''$).

The equation was derived by the Swiss mathematician Euler in 1736. The French mathematician Lagrange provided a much simplified proof in 1755.

Due to its importance, we give a brief proof here for those interested to have a taste.

The Euler Equation: Proof

Proof.

Suppose $x(t)$ is a solution to the above problem. For any function $a(t)$ satisfying $a(0) = a(T) = 0$, let $V(\epsilon) = \int_0^T F(t, x(t) + \epsilon a(t), x'(t) + \epsilon a'(t)) dt$. We know that $V(\epsilon)$ must be maximized at $\epsilon = 0$. Then, we have

$$\begin{aligned}
 0 &= \left. \frac{dV(\epsilon)}{d\epsilon} \right|_{\epsilon=0} = \int_0^T F_x(t, x(t), x'(t)) a(t) dt + \int_0^T F_{x'}(t, x(t), x'(t)) a'(t) dt \\
 &= \int_0^T F_x(t, x(t), x'(t)) a(t) dt + F_{x'}(t, x(t), x'(t)) a(t) \Big|_0^T - \int_0^T \frac{dF_{x'}(t, x(t), x'(t))}{dt} a(t) dt \\
 &= \int_0^T \left(F_x - \frac{dF_{x'}}{dt} \right) a(t) dt.
 \end{aligned} \tag{4}$$

The second line applies integration by parts and the third line uses the fact that $a(0) = a(T) = 0$. Since $a(t)$ is arbitrary, we have

$$F_x - \frac{dF_{x'}}{dt} = 0. \tag{5}$$



Discussions

The Euler equation can be written as

$$F_x - F_t - F_x x' - F_{x'} x'' = 0. \quad (6)$$

This is a 2nd-order differential equation of x , and thus we need two boundary conditions. The fundamental problem meets this requirement with $x(0) = x_0$ and $x(T) = x_T$.

However, we may encounter more general boundary conditions, e.g., one or both of T and $x(T)$ may be free, or $x(T)$ may be bounded by inequalities. To derive the related transversality conditions (TVC), we need the concept of variations.

Finally, for fixed T , the necessary and sufficient condition for the optimality of x is the Euler equation plus the TVC. And for T free, we could first solve for $V^*(T)$ given T , and then maximize over T .

(2) Optimal Control

The continued study of variational problems has led to the development of the more modern method of optimal control theory. In optimal control theory, the dynamic optimization problem is viewed as consisting of three (rather than two) types of variables. Aside from the time variable t and the state variable $x(t)$, consideration is given to a control variable $u(t)$. Indeed, it is the latter type of variable that gives optimal control theory its name and occupies the center of stage in this new approach to dynamic optimization.

To focus attention on the control variable implies that the state variable is relegated to a secondary status. This would be acceptable only if the decision on a control path $u(t)$ will, once given an initial condition on x , unambiguously determine a state-variable path $x(t)$ as a by-product. An optimal control problem hence contains an equation of motion:

$$\frac{dx(t)}{dt} = f(t, x(t), u(t)). \quad (7)$$

Optimal Control Problem

The optimal control problem corresponding to the calculus-of-variations problem (2) is as follows:

$$\max \quad V(u) = \int_0^T F(t, x(t), u(t)) dt, \quad (8)$$

$$\text{s.t.} \quad x'(t) = f(t, x(t), u(t)), \quad (9)$$

$$x(0) = x_0. \quad (10)$$

Now, u is the ultimate instrument of optimization. The calculus-of-variations problem (2) is a special case of the optimal control problem (8) when $x'(t) = u(t)$ and $x(T) = x_T$.

The single most significant development in optimal control theory is a first-order necessary condition known as *the Pontryagin maximum principle* developed in the 1940s-50s. The powerfulness of that principle lies in its ability to deal directly with certain constraints on the control variable (i.e., $u \in \mathcal{U}$, where $\mathcal{U}$ is some closed and bounded convex set.), thus it can deal with corner solutions, while the Euler equation applies only to “interior” solutions.

The Maximum Principle

If (x, u) is a solution to the problem (8) with u an interior solution of (11), then there exists λ , such that

$$\max_u H(t, x, u, \lambda), \quad \forall t \in [0, T], \quad (11)$$

$$x' = H_\lambda, \quad (12)$$

$$\lambda' = -H_x, \quad (13)$$

$$\lambda(T) = 0, \quad (14)$$

where

$$H(t, x, u, \lambda) = F(t, x, u) + \lambda f(t, x, u) \quad (15)$$

is called the Hamiltonian function and λ the Hamilton multiplier which, similar to the Lagrangian multiplier, is a *costate/auxillary variable*. Furthermore,

$$\frac{\partial H}{\partial u} = 0 \text{ and } \frac{\partial^2 H}{\partial u^2} \leq 0. \implies \frac{dH}{dt} = \frac{\partial H}{\partial t}. \quad (16)$$

(However, the optimal u need not be interior. The Hamiltonian H is even not required to be differentiable with respect to u .)

Proof.

For any feasible solution (y, v) , $y' = f(t, y, v)$ and $y(0) = x_0$. Then, we have

$$\begin{aligned}\int_0^T F(t, y, v) dt &= \int_0^T (F(t, y, v) + \lambda(f(t, y, v) - y')) dt = \int_0^T (F(t, y, v) + \lambda f(t, y, v)) dt - \int_0^T \lambda dy \\ &= \int_0^T (F(t, y, v) + \lambda f(t, y, v) + \lambda' y) dt - \lambda y|_0^T.\end{aligned}\quad (17)$$

Suppose (x, u) is an optimal solution, we can choose λ satisfying

$$\lambda' = -(F_x(t, x, u) + \lambda f_x(t, x, u)) \text{ and } \lambda(T) = 0. \quad (18)$$

Such λ , the solution to the above 1st-order differential equation with the terminal boundary condition, must exist if F, f are sufficient smooth. Now choose any function $g(t)$ defined on $[0, T]$ and $\forall \epsilon \in \mathbf{R}$. Let $u_\epsilon = u + \epsilon g$. Denote the feasible path corresponding to u_ϵ as x_ϵ . Then $x'_\epsilon = f(t, x_\epsilon, u_\epsilon)$ and $x_\epsilon(0) = x_0$. Similarly, x_ϵ must exist. Let

$$V(\epsilon) = \int_0^T F(t, x_\epsilon, u_\epsilon) dt = \int_0^T (F(t, x_\epsilon, u_\epsilon) + \lambda f(t, x_\epsilon, u_\epsilon) + \lambda' x_\epsilon) dt + \lambda(0)x_0. \quad (19)$$

We know that $V(\epsilon)$ should be maximized at $\epsilon = 0$. Then

$$0 = \left. \frac{dV(\epsilon)}{d\epsilon} \right|_{\epsilon=0} = \int_0^T \left((H_x + \lambda') \frac{\partial x_\epsilon}{\partial \epsilon} + H_u g \right) dt = \int_0^T H_u g dt. \quad (20)$$

Since g is arbitrary, we have $\frac{\partial H}{\partial u} = 0$. This in turn gives

$$H_x = F_x + \lambda f_x + H_u u_x = F_x + \lambda f_x = -\lambda', \quad (21)$$

$$\frac{dH}{dt} = \frac{\partial H}{\partial t} + \frac{\partial H}{\partial x} x' + \frac{\partial H}{\partial u} u' + \frac{\partial H}{\partial \lambda} \lambda' = \frac{\partial H}{\partial t} + H_x H_\lambda + H_\lambda (-H_x) = \frac{\partial H}{\partial t}. \quad (22)$$

□

Recall that the Calculus of Variations problem is equivalent to the Optimal Control problem with:

The Hamiltonian simplifies to:

For an interior solution, we have $\frac{\partial H}{\partial y} = 0$. This means:

Differentiating both side with respect to t , we get

This is just the Euler equation.

(3) Dynamic Programming

The American mathematician Richard Bellman pioneered the dynamic programming approach to the control problem stated in (8).

The most important distinguishing characteristics of this approach are two:

- ◇ First, it embeds the given control problem in a family of control problems, with the consequence that in solving the given problem, we are actually solving the entire family of problems.
- ◇ Second, for each member of this family of problems, primary attention is focused on the optimal value of the functional, V^* , rather than on the properties of the optimal state path $x^*(t)$ (as in the calculus of variations) or the optimal control path $u^*(t)$ (as in optimal control theory). In fact, an optimal value function—assigning an optimal value to each individual member of this family of problems—is used as a characterization of the solution.

An Example to Illustrate Dynamic Programming

We illustrate the use of backward induction to solve a simple finite-horizon dynamic optimization problem, that is, the consumption-savings problem.

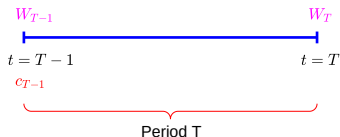
The Consumption-Savings Problem

An agent has initial wealth $\bar{w} > 0$ at $t = 0$. She has a planning horizon of T periods, where T is a fixed strictly positive integer. If she has wealth W_{t-1} at the beginning of period t and chooses to consume c_{t-1} ($0 \leq c_{t-1} \leq W_{t-1}$) that period, then her wealth at the end of period t or the beginning of the next period $t + 1$ is

$W_t = (W_{t-1} - c_{t-1})(1 + r)$, where r is the interest rate, and she derives utility $U(c_{t-1}) = \sqrt{c_{t-1}}$ from consumption in period t . Her objective is to maximize her utility over the entire horizon T . To simplify notation, we write $k = 1 + r$. Mathematically, she solves

$$\begin{aligned}
 V_0^*(W_0 = \bar{w}) = & \max_{\{c_t\}_{t=0}^{T-1}} \sum_{t=0}^{T-1} \sqrt{c_t}, \\
 \text{s.t. } & W_{t+1} = k(W_t - c_t), \quad t = 0, 1, \dots, T-1, \\
 & c_t \in [0, W_t], \quad t = 0, 1, \dots, T-1.
 \end{aligned} \tag{27}$$

The Last Period



For the last period, she solves

$$\begin{aligned} \max_{c_{T-1} \in [0, W_{T-1}]} \quad & \sqrt{c_{T-1}}, \\ \text{s.t.} \quad & W_T = k(W_{T-1} - c_{T-1}). \end{aligned} \quad (28)$$

Since there is no bequest motive, the optimal policy is to consume everything as the utility function is strictly increasing. Thus, the unique optimal consumption strategy for the one-period problem is

$$c_{T-1}^*(W_{T-1}) = W_{T-1}, \quad \forall W_{T-1} \geq 0. \quad (29)$$

Her value function is

$$V_{T-1}^*(W_{T-1}) = \sqrt{W_{T-1}}. \quad (30)$$

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A digression

The multi-period problem can actually be solved using our familiar method of slack control.
For the last two periods, she solves

$$\max_{c_{T-1}, c_{T-2}, W_T, W_{T-1}} \sqrt{c_{T-2}} + \sqrt{c_{T-1}}, \quad (34)$$

$$\text{s.t. } W_T = k(W_{T-1} - c_{T-1}), \quad (35)$$

$$W_{T-1} = k(W_{T-2} - c_{T-2}), \quad (36)$$

$$c_{T-1} - W_{T-1} \leq 0, \quad (37)$$

$$c_{T-2} - W_{T-2} \leq 0, \quad (38)$$

$$-c_{T-1} \leq 0, \quad (39)$$

$$-c_{T-2} \leq 0, \quad (40)$$

with $W_{T-2} > 0$.

Form the Lagrangian:

$$\mathcal{L} = \sqrt{c_{T-2}} + \sqrt{c_{T-1}} + \sum_{i=1}^2 \alpha_i (W_{T-i+1} - k(W_{T-i} - c_{T-i})) + \sum_{i=1}^2 \beta_i (c_{T-i} - W_{T-i}) - \sum_{i=1}^2 \gamma_i c_{T-i}. \quad (41)$$

We have $2^4 = 16$ cases to discuss. However, we can simplify the analysis using economic principles.

First, note that the investor has no incentive to leave bequest. Thus, we must have (37) in equality. And no more than one of (38)-(40) could also be in equality (by constraint qualifications).

Second,

- ◇ if (38) holds as an equality, we have the optimal solution: $c_{T-2}^* = W_{T-2}$, $c_{T-1}^* = 0$ and the value function equals $\sqrt{W_{T-2}}$;
- ◇ if (39) holds as an equality, we also have the optimal solution: $c_{T-2}^* = W_{T-2}$, $c_{T-1}^* = 0$ and the value function equals $\sqrt{W_{T-2}}$;
- ◇ if (40) holds as an equality, we have the optimal solution: $c_{T-2}^* = 0$, $c_{T-1}^* = kW_{T-2}$ and the value function equals $\sqrt{kW_{T-2}} > \sqrt{W_{T-2}}$ since $k > 1$.

$\beta_2 = \gamma_1 = \gamma_2 = 0$. The F.O.C.s for c_{T-1} , c_{T-2} , W_T , W_{T-1} , α_1 , α_2 , and β_1 are

$$\frac{1}{2\sqrt{c_{T-1}}} + k\alpha_1 + \beta_1 = 0,$$

$$\frac{1}{2\sqrt{c_{T-2}}} + k\alpha_2 = 0,$$

$$\alpha_1 = 0,$$

$$-k\alpha_1 + \alpha_2 - \beta_1 = 0,$$

$$W_T = k(W_{T-1} - c_{T-1}),$$

$$W_{T-1} = k(W_{T-2} - C_{T-2}),$$

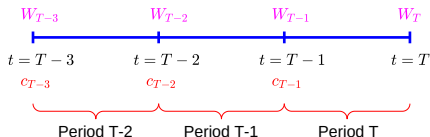
$$c_{T-1} - w_{T-1} = 0.$$

We have the optimal solution: $c_{T-2}^* = \frac{W_{T-2}}{1+k}$, $c_{T-1}^* = \frac{k^2 W_{T-2}}{1+k}$, and the value function equals $\sqrt{1+k} \sqrt{W_{T-2}} > \sqrt{k W_{T-2}}$. Thus, this case gives the optimal consumption strategy.

Although we achieve the same results, the method quickly becomes cumbersome when the number of time periods increases!

The Last Three Periods

Question 1



Write down the optimization problem for the last three periods, and then solve for the optimal consumption strategy $c_{T-3}^*(W_{T-3})$ and the value function $V_{T-3}^*(W_{T-3})$.

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The General Solution

We see that there is a pattern. (If you do not yet believe it, work out the cases for the last four, five, periods.) We guess that the general solution for the problem starting at time t ($0 \leq T \leq T - 1$) is as follows.

$$c_t^*(W_t) = \frac{W_t}{1 + k + \dots + k^{T-t-1}}, \forall W_t \geq 0. \quad (45)$$

$$V_t^*(W_t) = \sqrt{1 + k + \dots + k^{T-t-1}} \sqrt{W_t}. \quad (46)$$

Therefore, the solution to the original problem is

$$c_0^*(\bar{w}) = \frac{\bar{w}}{1 + k + \dots + k^{T-1}}, \forall \bar{w} \geq 0. \quad (47)$$

$$V_0^*(\bar{w}) = \sqrt{1 + k + \dots + k^{T-1}} \sqrt{\bar{w}}. \quad (48)$$

Proof.

At time point $t \in \{0, 1, \dots, T - 1\}$, which is the beginning of period $t + 1$, the agent solves:

where $V_{t+1}^*(x) = \sqrt{1 + k + \dots + k^{T-t-2}}\sqrt{x}$, $\forall x \geq 0$ and $t \in \{0, 1, \dots, T-2\}$, and $V_T^*(x) \equiv 0$, $\forall x \geq 0$.

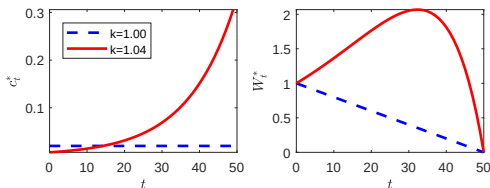
$$c_t^*(W_t) = \frac{W_t}{1+k+\dots+k^{T-t-1}}, \forall W_t \geq 0. \quad (50)$$

This completes the induction.

Economical Implications

How do the parameters, namely, $\bar{w}$, k , and T , influence the solution?

Draw the trajectories of optimal consumption growth and wealth accumulation. Say, choose $\bar{w} = 1$ and $T = 50$.



Subjective Discounting

Question 2

The agent is now impatient and hence reluctant to delay consumption into the future. She imposes a subjective discount on future consumption by solving:

$$V_0^*(W_0 = \bar{w}) = \max_{\{c_t\}_{t=0}^{T-1}} \sum_{t=0}^{T-1} \rho^t \sqrt{c_t}, \quad (52)$$

$$\text{s.t. } W_{t+1} = k(W_t - c_t), \quad t = 0, 1, \dots, T-1,$$

where $0 < \rho \leq 1$.

Find her optimal consumption strategy and value function.

Question 2: Answer

For the last period, she solves

$$\begin{aligned} \max_{c_{T-1} \in [0, W_{T-1}]} \quad & \rho^0 \sqrt{c_{T-1}}, \\ \text{s.t.} \quad & W_T = k(W_{T-1} - c_{T-1}). \end{aligned} \quad (53)$$

Her optimal consumption strategy and value function are

$$c_{T-1}^*(W_{T-1}) = W_{T-1}, \forall W_{T-1} \geq 0. \quad (54)$$

$$V_{T-1}^*(W_{T-1}) = \sqrt{W_{T-1}}. \quad (55)$$

For the last two periods, she solves

$$\max_{c_{T-2} \in [0, W_{T-2}]} \quad \sqrt{c_{T-2}} + \rho V_{T-1}^*(k(W_{T-2} - c_{T-2})). \quad (56)$$

Her optimal consumption strategy and value function are

$$c_{T-2}^*(W_{T-2}) = \frac{W_{T-2}}{1 + k\rho^2}, \forall W_{T-2} \geq 0. \quad (57)$$

$$V_{T-2}^*(W_{T-2}) = \sqrt{1 + k\rho^2} \sqrt{W_{T-2}}. \quad (58)$$

For the last three periods, she solves

$$\max_{c_{T-3} \in [0, W_{T-3}]} \sqrt{c_{T-3}} + \rho V_{T-2}^*(k(W_{T-3} - c_{T-3})), \quad (59)$$

Her optimal consumption strategy and value function are

$$c_{T-3}^*(W_{T-3}) = \frac{W_{T-3}}{1 + k\rho^2 + (k\rho^2)^2}, \quad \forall W_{T-3} \geq 0. \quad (60)$$

$$V_{T-3}^*(W_{T-3}) = \sqrt{1 + k\rho^2 + (k\rho^2)^2} \sqrt{W_{T-3}}. \quad (61)$$

Generally, at time point $t \in \{0, 1, \dots, T - 1\}$, which is the beginning of period $t + 1$, she solves:

$$\max_{c_t \in [0, W_t]} \sqrt{c_t} + \rho V_{t+1}^*(k(W_t - c_t)), \quad (62)$$

where we define $V_T^*(x) \equiv 0, \forall x \geq 0$, and suppose $V_{t+1}^*(x) = \sqrt{1 + k\rho^2 + \dots + (k\rho^2)^{T-t-2}}\sqrt{x}$, where $x \geq 0$ and $t \in \{0, 1, \dots, T-2\}$.

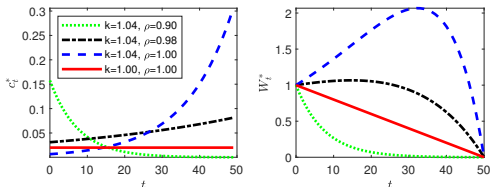
This is a strictly convex optimization problem. We compute her optimal consumption strategy and the value function as

$$c_t^*(W_t) = \frac{W_t}{1 + k\rho^2 + \dots + (k\rho^2)^{T-t-1}}, \forall W_t \geq 0. \quad (63)$$

$$V_t^*(W_t) = \sqrt{1 + k\rho^2 + \dots + (k\rho^2)^{T-t-1}} \sqrt{W_t}. \quad (64)$$

We have achieved the results by induction.

Choose $\bar{w} = 1$ and $T = 50$.



Bequest

Question 3

The agent now derives utility $\sqrt{W_T}$ from leaving a bequest W_T at time T . She solves:

$$V_0^*(W_0 = \bar{w}) = \max_{\{c_t\}_{t=0}^{T-1}} \sum_{t=0}^{T-1} \sqrt{c_t} + \sqrt{W_T}, \quad (65)$$

$$\text{s.t. } W_{t+1} = k(W_t - c_t), \quad t = 0, 1, \dots, T-1.$$

Find her optimal consumption strategy and value function.

Question 3: Answer

For the last period, she solves

$$\begin{aligned} \max_{c_{T-1} \in [0, W_{T-1}]} \quad & \sqrt{c_{T-1}} + \sqrt{W_T}, \\ \text{s.t.} \quad & W_T = k(W_{T-1} - c_{T-1}). \end{aligned} \quad (66)$$

Her optimal consumption strategy and value function are

$$c_{T-1}^*(W_{T-1}) = \frac{W_{T-1}}{1+k}, \quad \forall W_{T-1} \geq 0. \quad (67)$$

$$V_{T-1}^*(W_{T-1}) = \sqrt{1+k} \sqrt{W_{T-1}}. \quad (68)$$

For the last two periods, she solves

$$\max_{c_{T-2} \in [0, W_{T-2}]} \quad \sqrt{c_{T-2}} + V_{T-1}^*(k(W_{T-2} - c_{T-2})). \quad (69)$$

Her optimal consumption strategy and value function are

$$c_{T-2}^*(W_{T-2}) = \frac{W_{T-2}}{1+k+k^2}, \quad \forall W_{T-2} \geq 0. \quad (70)$$

$$V_{T-2}^*(W_{T-2}) = \sqrt{1+k+k^2} \sqrt{W_{T-2}}. \quad (71)$$

10. *Journal of the American Statistical Association*, 1997, 92, 1009-1014.

$$\max_{c_t \in [0, W_t]} \sqrt{c_t} + V_{t+1}^*(k(W_t - c_t)). \quad (72)$$

Suppose $V_{t+1}^*(x) = \sqrt{1 + k + \dots + k^{T-t-1}} \sqrt{x}, \forall x \geq 0$.

This is a strictly convex optimization problem. We compute her optimal consumption strategy and the value function as

$$c_t^*(W_t) = \frac{W_t}{1+k+\dots+k^{T-t}}, \forall W_t \geq 0. \quad (73)$$

$$V_t^*(W_t) = \sqrt{1 + k + \dots + k^{T-t}} \sqrt{W_t}. \quad (74)$$

We have achieved the results by induction.

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Continuous-Time Model

- ◇ For more problems of dynamic investment, it is convenient to think of time as a continuous variable, other than a discrete succession of periods.
- ◇ We can think of continuous time as the limit when we take discrete periods of length Δt , and let $\Delta t \rightarrow 0$.
- ◇ We use the consumption-savings problem as an example to illustrate such a transition.

The investor solves:

Or equivalently, she solves:

where $\Delta t = 1$.

Now, let $\Delta t \rightarrow 0$. We first see that the objective function converges to the following integral.

$$\int_0^T \sqrt{c_t} dt. \quad (80)$$

We rewrite the state equation as:

$$\frac{W_{t+\Delta t} - W_t}{\Delta t} = rW_t - c_t - rc_t\Delta t, \quad t = 0, \Delta t, \dots, T - \Delta t. \quad (81)$$

Then,

$$\frac{dW_t}{dt} = rW_t - c_t, \quad t \in [0, T]. \quad (82)$$

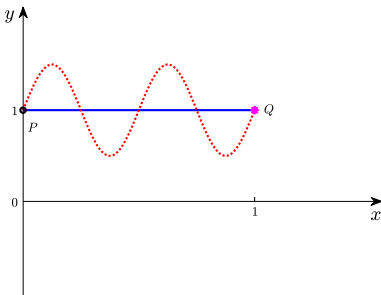
In continuous time, the investor solves:

$$\begin{aligned} V_0^*(W_0) = & \max_{c_t \geq 0, t \in [0, T]} \int_0^T \sqrt{c_t} dt, \\ \text{s.t. } & \frac{dW_t}{dt} = rW_t - c_t, \\ & W_0 = \bar{w}. \end{aligned} \quad (83)$$

Note that c_t becomes the rate of consumption, thus $c_t dt \leq W_t$ is always satisfied as long as $W_t > 0$, which is in turn guaranteed if $\bar{w} > 0$ and c_t does not grow too fast.

In practice, we often assume that the optimal c_t is interior, and once we get the solution, we can go back to verify that $\forall t \in [0, T], c_t \geq 0$ and $W_t \geq 0$.

And then we will come back to the consumption-savings problem and give the Principle of Optimality of dynamic programming in continuous time.



$$\begin{aligned} \max \quad & V(y) = \int_0^1 -\sqrt{1 + (y')^2} dx, \\ \text{s.t.} \quad & y(0) = 1 \text{ and } y(1) = 1. \end{aligned} \quad (84)$$

$$F_x = 0, F_y = 0, \text{ and } F_{y'} = -\frac{y'}{\sqrt{1 + (y')^2}}. \quad (85)$$

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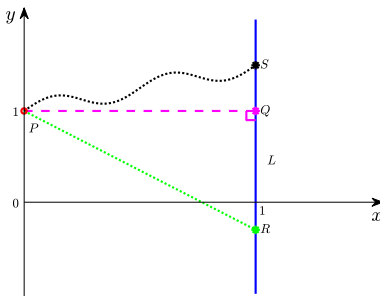
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Step ii

The optimal straight line is uniquely determined by the two end points.
We obtain the solution:

$$y = 1, \quad 0 \leq x \leq 1. \quad (89)$$

Find the curve with the shortest distance from the given point $P : (0, 1)$ to the given vertical line $L : x = 1$. We know that the solution is the line segment from P to $Q : (1, 1)$, that is, $y = 1, 0 \leq x \leq 1$.



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$$\begin{aligned} \max \quad & V(y) = \int_0^1 -\sqrt{1 + (y')^2} dx, \\ \text{s.t.} \quad & y(0) = 1 \text{ and } y(1) \text{ free.} \end{aligned} \quad (90)$$

$$F_x = 0, F_y = 0, \text{ and } F_{y'} = -\frac{y'}{\sqrt{1 + (y')^2}}. \quad (91)$$

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- 1 The first condition is that the shortest path be a straight line: one cannot make the path shorter by deviating from it temporarily but eventually returning to it. This is the implication of the Euler equation.
- 2 A straight line from P to L could still be infinitely long. Thus very bad decisions are still possible.
- 3 The second condition requires the shortest path to be perpendicular to L . It ensures that one cannot make the path shorter by deviating from it and never returning to it.
- 4 The condition of perpendicularity in this example and similar conditions on end points in other problems are called transversality conditions² in dynamic optimization theory.

²The term was first introduced by Adolf Kneser in 1900. See Kneser, A. 1900. *Lehrbuch der Variationsrechnung*. Braunschweig: F. Vieweg und Sohn.

By Optimal Control

Consider again the problem of finding the curve with the shortest distance from the given point $P : (0, 1)$ to the given vertical line $L : x = 1$, with the known solution: the line segment from P to $Q : (1, 1)$, that is, $y = 1, 0 \leq x \leq 1$.

We first formulate the problem in the context of optimal control.

$$\begin{aligned} \max \quad & V(u) = \int_0^1 -\sqrt{1+u^2} dx, \\ \text{s.t.} \quad & y' = u. \\ & y(0) = 1 \text{ and } y(1) \text{ free.} \end{aligned} \tag{97}$$

We have $F(x, y, u) = -\sqrt{1+u^2}$.

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principle".

Step ii

The equation of motion of the costate variable in (13) yields:

$$\lambda' = -H_y = 0. \quad (101)$$

Thus, $\lambda = \text{constant}$.

The transversality condition in (14) requires that $\lambda(1) = 0$, which then suffices to determine the constant. And we have

$$\lambda(x) = 0, \quad \forall x \in [0, 1]. \quad (102)$$

$$u(x) = 0, \quad \forall x \in [0, 1]. \quad (103)$$

Step iii

The equation of motion of the state variable in (12) yields:

$$y' = H_\lambda = u = 0. \quad (104)$$

Note the symmetry in the two equations of motion for the state and costate variables. Also note that the equation of motion for the state variable is just a restatement of the state-control relation in (9).

Making use of the initial condition $y(0) = 1$, we obtain the optimal state path:

$$y(x) = 1, \quad \forall x \in [0, 1]. \quad (105)$$

Question 4

We restate the optimal control problem in (8)-(10) below.

$$\begin{aligned} \max \quad & \int_0^T F(t, x(t), u(t)) dt, \\ \text{s.t.} \quad & x'(t) = f(t, x(t), u(t)), \quad t \in [0, T] \\ & x(0) = x_0. \end{aligned}$$

It can be written in discrete time as follows.

$$\begin{aligned} \max \quad & \sum_{t=0}^{T-1} F(t, x(t), u(t)), \\ \text{s.t.} \quad & x_{t+1} - x_t = f(t, x(t), u(t)), \quad t = 0, 1, \dots, T-1, \\ & x(0) = x_0. \end{aligned} \tag{106}$$

Show the transition from discrete- to continuous-time.

Question 4: Answer

The investor solves:

$$\begin{aligned} \max \quad & \sum_{i=0}^{T/\Delta t - 1} F(i\Delta t, x(i\Delta t), u(i\Delta t))\Delta t, \\ \text{s.t.} \quad & x_{t+\Delta t} - x_t = f(t, x(t), u(t))\Delta t, \quad t = 0, \Delta t, \dots, T - \Delta t, \\ & x(0) = x_0. \end{aligned} \tag{107}$$

where $\Delta t = 1$.

Letting $\Delta t \rightarrow 0$, we get the continuous-time limit.

Consider again the following problem.

$$x(0) = x_0. \quad (110)$$

For any $s \in [0, T]$, define the subproblem:

$$x(s) = x_s. \quad (113)$$

where $V(s, x_s)$ is maximum value of the objective functional.
Obviously, $V(T, x_T) = 0$.

Suppose x is an optimal solution to (111), then for $s \in [0, T)$ and a sufficiently small $\Delta s > 0$,

$$\begin{aligned}
 V(s, x_s) &= \max_u \int_s^T F(t, x(t), u(t)) dt \\
 &= \max_u \left(\int_s^{s+\Delta s} F(t, x(t), u(t)) dt + \int_{s+\Delta s}^T F(t, x(t), u(t)) dt \right) \\
 &= \max_u \left(\int_s^{s+\Delta s} F(t, x(t), u(t)) dt + V(s + \Delta s, x(s + \Delta s)) \right) \\
 &= \max_u (F(s, x(s), u(s))\Delta s + V(s, x_s) + V_s\Delta s + V_x\Delta x(s)).
 \end{aligned}$$

$$\implies 0 = \max_u (F(s, x(s), u(s))\Delta s + V_s\Delta s + V_x\Delta x(s)). \quad (114)$$

Dividing both sides by Δs and letting $\Delta s \rightarrow 0$, we have

This is the **HJB Equation**, a partial differential equation (PDE) with the boundary condition $V(T, x_T) = 0$. This is also we call the **Principle of Optimality**.

Note that the maximand on the right hand side is just the Hamiltonian $H = F + \lambda f$ with $\lambda = V_x$ as the multiplier.

Furthermore, if the optimal u is interior, then

This is just $H_u = 0$. We can express u as $u(s, x_s)$ to eliminate u in (115).

Intuition for the Hamiltonian

Let's take another look at the Hamiltonian.

$$\max_u (F(s, x(s), u(s)) + V_x f(s, x(s), u(s))) . \quad (117)$$

We know that $u(s)$ impacts not only the immediate reward $F(s, x(s), u(s))$ but also the evolution of the state variable x and hence all future rewards. However, [the above Hamiltonian function reduces this complicated dynamic decision to a simple static one](#). Note that all the variables are fixed at time t . This provides a great conceptual merit.

The reason is that the effect of u on the state evolution is summarized by f ($\Delta x = f \Delta s$), and the effect of the state x on the value function V , which comprises all future rewards, is measured, to the 1st order, at V_x . Therefore, the last term $V_x f$ captures the total effect of u on future rewards.

The fact $\lambda = V_x$ clarifies the economic meaning of the costate variable λ . It also explains the transversality condition $\lambda(T) = 0$, as $V(T, x_T) = 0$ and thus $V_x(T, x_T) = 0$.

Then,

$$\begin{aligned}
 -\lambda' &= -dV_x/ds = -V_{xs} - V_{xx}x' = -V_{sx} - V_{xx}f \\
 &= F_x + F_u u_x + V_{xx}f + V_x f_x + V_x f_u u_x - V_{xx}f \\
 &= F_x + V_x f_x + (F_u + V_x f_u)u_x \\
 &= F_x + V_x f_x + H_u u_x \\
 &= F_x + V_x f_x \\
 &= H_x.
 \end{aligned}
 \tag{119}$$

The PDE in (118) can be solved analytically in certain cases, while numerical solutions are generally entailed. Since we are not mathematicians, a practical way is to look up a reference book which lists known solutions to various PDEs. If unlucky, we turn to educated guessing of solutions. If still unlucky, we either go back to revise the model or turn to brute force numerical methods.

Shortest Distance By Dynamic Programming

Consider again the problem of finding the curve with the shortest distance from the given point $P : (0, 1)$ to the given vertical line $L : x = 1$, with the known solution: the line segment from P to $Q : (1, 1)$, that is, $y = 1, 0 \leq x \leq 1$.

We first formulate the problem in the context of dynamic programming.

$$\begin{aligned} V(t, y_t) = \max & \int_t^1 -\sqrt{1+u^2} dx, \\ \text{s.t. } & y' = u, \\ & y(t) = y_t \text{ and } y(1) \text{ free,} \end{aligned} \quad (120)$$

where $t \in [0, 1]$. We have $F(x, y, u) = -\sqrt{1 + u^2}$.

The HJB equation in (115) implies:

$$-V_t = \max_u \left(-\sqrt{1+u^2} + V_y u \right), \quad (121)$$

with the boundary condition $V(1, y(1)) = 0$. The optimal u is computed at:

$$u = \frac{V_y}{\sqrt{1 - V_y^2}}. \quad (122)$$

Inserting this solution back into (121) and rearranging yield:

$$V_t - \sqrt{1 - V_y^2} = 0. \quad (123)$$

$$V(t, y(t)) = ct + d, \quad (124)$$
$$V_t - \sqrt{1 - V_y^2} = c - 1 = 0, \quad (125)$$

This gives us $c = 1$, $d = -1$, and $V(t, y(t)) = t - 1$. Moreover,

Hence, y is a constant, and we decide $y = 1$ ($t \leq x \leq 1$) by the initial condition $y(0) = 1$. Letting $t = 0$, we get the solution to the original problem: $y = 1$ ($0 \leq x \leq 1$).

A summary

Notice that the state variable (y , through the Euler Equation), the control variable (u , through the Maximum Principle), and the value function (V , through the Principle of Optimality) stand at the center of the stage for calculus of variations, optimal control, and dynamic programming, respectively.

We will solve it using dynamic programming, which is quite amenable to extensions that combine time and uncertainty.⁴

At time $t \in [0, T]$, the investor solves:

$$\begin{aligned} \text{s.t. } \quad & \frac{dW_s}{ds} = rW_s - c_s, \\ & W_t \text{ is fixed and positive, and } W(T) \text{ free,} \end{aligned} \quad (129)$$

where we have temporarily ignored the constraint $c_s \geq 0$ (and $W_s \geq 0$). And we can come back to verify it when we will have obtained the optimal solution.

⁴A more advanced version of optimal control, that is, stochastic optimal control, can suit the same purpose. We leave the fun to your future exploration.

—114—

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99

$$1 - (rb(t) + 1)^{-1}$$

$$d \mid n(rh(t) + 1)$$

$$c^T \quad c^T$$

Math. 1 (7) = 2.000, Math. 2 (7) = 2.000, Math. 3 (7) = 2.000

$$W_t = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{W_t}{\sigma}\right)^2\right)$$

Wealth Accumulation

Finally, using the optimal solution of c_t , we have

$$\frac{dW_t}{dt} = rW_t - c_t = \left(r - \frac{1}{(e^{r(T-t)} - 1)/r} \right) W_t, \quad (145)$$

$$\Rightarrow \int_0^t d \ln W_t = \int_0^t r dt + \int_0^t d \ln(1 - e^{-r(T-t)}), \quad (146)$$

with the initial condition $W_0 = \bar{w}$. Solving it yields:

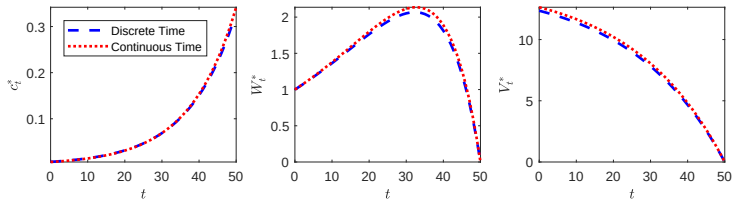
$$W_t = \frac{1 - e^{-r(T-t)}}{1 - e^{-rT}} e^{rt} \bar{w}, \quad c_t = \frac{re^{2rt}}{e^{rT} - 1} \bar{w}. \quad (147)$$

Then,

$$\lim_{t \rightarrow T} W_t = 0, \quad \lim_{t \rightarrow T} c_t = \frac{re^{2rT}}{e^{rT} - 1} \bar{w}, \quad (148)$$

$$\lim_{r \rightarrow 0} W_t = \frac{T-t}{T} \bar{w}, \quad \lim_{r \rightarrow 0} c_t = \frac{\bar{w}}{T}. \quad (149)$$

Choose $\bar{w} = 1$, $r = 0.04$, and $T = 50$.



(Here, to put things on equal footing, we plot $V/2$ instead of V under the continuous-time case.)

Question 5: Subjective Discounting

The optimal control problem with subjective discounting in discrete time is as below.

$$\begin{aligned} \max \quad & \sum_{t=0}^{T-1} \tilde{\rho}^t F(t, x(t), u(t)), \\ \text{s.t.} \quad & x_{t+1} - x_t = f(t, x(t), u(t)), \quad t = 0, 1, \dots, T-1, \\ & x(0) = x_0. \end{aligned} \tag{150}$$

It can be rewritten as:

$$\begin{aligned} \max \quad & \sum_{i=0}^{T/\Delta t - 1} (1 - \rho\Delta t)^i F(i\Delta t, x(i\Delta t), u(i\Delta t))\Delta t, \\ \text{s.t.} \quad & x_{t+\Delta t} - x_t = f(t, x(t), u(t))\Delta t, \quad t = 0, \Delta t, \dots, T - \Delta t, \\ & x(0) = x_0. \end{aligned} \tag{151}$$

where $\rho = 1 - \tilde{\rho}$ and $\Delta t = 1$.

Show the transition from discrete- to continuous-time.

$$\text{s.t. } x'(t) = f(t, x(t), u(t)), \quad (154)$$

HJB Equation with Subjective Discounting

For any $s \in [0, T]$, define the subproblem:

$$V(s, x_s) = \max \int_s^T e^{-\rho(t-s)} F(t, x(t), u(t)) dt, \quad (156)$$

$$s.t. \quad x'(t) = f(t, x(t), u(t)), \quad (157)$$

$$x(s) = x_s, \quad (158)$$

where $V(s, x_s)$ is maximum value of the objective functional.
Obviously, $V(T, x_T) = 0$.

$$V(s, x_s)$$

$$\Rightarrow 0 = \max_u (F(s, x(s), u(s))\Delta s + V_s\Delta s + V_x\Delta x(s) - \rho V(s, x_s)\Delta s). \quad (159)$$

$$\rho V = \max_u (V_s + F(s, x(s), u(s)) + V_x f(s, x(s), u(s))). \quad (160)$$

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For this problem,

$$f(s, W_s, c_s) = rW_s - c_s. \quad (164)$$

$$\rho V = \max_{c_t} \left(V_t + 2\sqrt{c_t} + V_W(rW_t - c_t) \right), \quad (165)$$
$$c_t = V_W^{-2}. \quad (166)$$
$$\rho V - V_t - V_W^{-1} - rW_t V_W = 0. \quad (167)$$

Again, we guess a solution of the following form:

$$V(t, W_t) = \sqrt{h(t)}\sqrt{W_t}, \quad (168)$$

where $h(t)$ is a function of t . We have:

$$\left(\rho \sqrt{h(t)} - \frac{h'(t)}{2\sqrt{h(t)}} - \frac{2}{\sqrt{h(t)}} - \frac{r\sqrt{h(t)}}{2} \right) \sqrt{W_t} = 0. \quad (169)$$

This is true for any $W_t \geq 0$. Thus

$$\frac{h'(t)}{2\sqrt{h(t)}} + \frac{2}{\sqrt{h(t)}} + \frac{(r-2\rho)\sqrt{h(t)}}{2} = 0, \quad (170)$$

with the terminal condition $h(T) = 0$.

Observing that if we replace $r - 2\rho$ by r , then we get exactly the same equation as that under the case without subjective discounting. Therefore, we can directly write down the solution as follows.

$$h(t) = 4 \left(e^{(r-2\rho)(T-t)} - 1 \right) / (r - 2\rho), \quad (171)$$

$$C_t = \frac{W_t}{(e^{(r-2\rho)(T-t)} - 1)/(r - 2\rho)}, \quad (172)$$

$$V(t, W_t) = 2\sqrt{(e^{(r-2\rho)(T-t)} - 1)/(r - 2\rho)}\sqrt{W_t}. \quad (173)$$

Why is there a “2” in the adjustment to r ?

Recall that in the discrete-time case, we replace k by $k\tilde{\rho}^2$. For a small time interval Δt , $(1 + r\Delta t)(1 - \rho\Delta t)^2 = 1 + (r - 2\rho)\Delta t + o(\Delta t)$.

Wealth Accumulation

Finally, using the optimal solution of c_t , we have

$$\frac{dW_t}{dt} = rW_t - c_t = \left(r - \frac{1}{(e^{(r-2\rho)(T-t)} - 1)/(r-2\rho)} \right) W_t, \quad (174)$$

$$\Rightarrow \int_0^t d \ln W_t = \int_0^t r dt + \int_0^t d \ln(1 - e^{-(r-2\rho)(T-t)}), \quad (175)$$

with the initial condition $W_0 = \bar{w}$. Solving it yields:

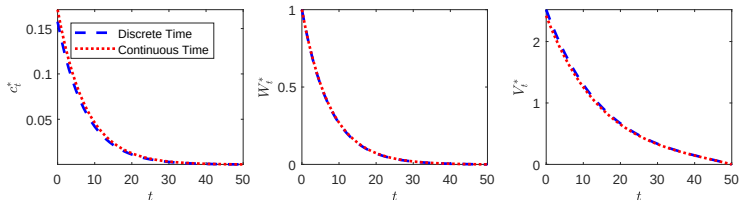
$$W_t = \frac{1 - e^{-(r-2\rho)(T-t)}}{1 - e^{-(r-2\rho)T}} e^{rt} \bar{w}, \quad c_t = \frac{(r-2\rho)e^{2(r-\rho)t}}{e^{(r-2\rho)T} - 1} \bar{w}. \quad (176)$$

Then,

$$\lim_{t \rightarrow T} W_t = 0, \quad \lim_{t \rightarrow T} c_t = \frac{(r-2\rho)e^{2(r-\rho)T}}{e^{(r-2\rho)T} - 1} \bar{w}, \quad (177)$$

$$\lim_{r \rightarrow 2\rho} W_t = \frac{T-t}{T} e^{2\rho t} \bar{w}, \quad \lim_{r \rightarrow 2\rho} c_t = \frac{e^{2\rho t} \bar{w}}{T}. \quad (178)$$

Choose $\bar{w} = 1$, $r = 0.04$, $\tilde{\rho} = 0.90$, $\rho = -\ln(0.90) = 0.1054$, and $T = 50$.



(Here, to put things on equal footing, we plot $V/2$ instead of V under the continuous-time case.)

HJB Equation with Bequest

For any $s \in [0, T]$, define the subproblem:

$$V(s, x_s) = \max \left(\int_s^T F(t, x(t), u(t)) dt + B(T, x(T)) \right), \quad (179)$$

$$s.t. \quad x'(t) = f(t, x(t), u(t)), \quad (180)$$

$$x(s) = x_s. \quad (181)$$

where $B(T, x(T))$ captures the motivation to leave a bequest.
Obviously, $V(T, x_T) = B(T, x_T)$.

Suppose x is an optimal solution to (179), then for $s \in [0, T)$ and a sufficiently small $\Delta s > 0$,

$$\begin{aligned} V(s, x_s) &= \max_u \left(\int_s^T F(t, x(t), u(t)) dt + B(T, x(T)) \right) \\ &= \max_u \left(\int_s^{s+\Delta s} F(t, x(t), u(t)) dt + \int_{s+\Delta s}^T F(t, x(t), u(t)) dt + B(T, x(T)) \right) \\ &= \max_u \left(\int_s^{s+\Delta s} F(t, x(t), u(t)) dt + V(s + \Delta s, x(s + \Delta s)) \right) \\ &= \max_u \left(F(s, x(s), u(s)) \Delta s + V(s, x_s) + V_s \Delta s + V_x \Delta x(s) \right). \end{aligned}$$

$$\Rightarrow 0 = \max_u (F(s, x(s), u(s))\Delta s + V_s\Delta s + V_x\Delta x(s)). \quad (182)$$

Dividing both sides by Δs and letting $\Delta s \rightarrow 0$, we have

$$-V_s = \max_u (F(s, x(s), u(s)) + V_x f(s, x(s), u(s))). \quad (183)$$

This is the **HJB Equation** with bequest. It is the same as that without bequest. However, notice that the terminal condition is now different.

[illegible]

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For this problem,

$$f(s, W_s, c_s) = rW_s - c_s. \quad (187)$$

Thus, the HJB equation is

$$-V_t = \max_{c_t} \left(2\sqrt{c_t} + V_W(rW_t - c_t) \right), \quad (188)$$

with the boundary condition $V(T, W_T) = 2\sqrt{W_T}$. The optimal c_t is computed at:

$$C_t = \frac{1}{V_W^2}. \quad (189)$$

Inserting this solution back into (188) and rearranging yield:

$$V_t + \frac{1}{V_W} + rW_t V_W = 0. \quad (190)$$

$$V(t, W_t) = \sqrt{h(t)}\sqrt{W_t}, \quad (191)$$
$$\left(\frac{h'(t)}{2\sqrt{h(t)}} + \frac{2}{\sqrt{h(t)}} + \frac{r\sqrt{h(t)}}{2} \right) \sqrt{W_t} = 0. \quad (192)$$
$$\frac{h'(t)}{2\sqrt{h(t)}} + \frac{2}{\sqrt{h(t)}} + \frac{r\sqrt{h(t)}}{2} = 0, \quad (193)$$

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We get after rearranging:

$$h'(t) + 4 + rh(t) = 0, \quad (194)$$

$$\Rightarrow \int_t^T d \ln(rh(t) + 4) = \int_t^T -r dt, \quad (195)$$

with the terminal condition $h(T) = 4$. Solving it yields:

$$h(t) = 4 \left((1 + r) e^{r(T-t)} - 1 \right) / r, \quad (196)$$

$$c_t = \frac{W_t}{((1+r)e^{r(T-t)} - 1)/r}, \quad (197)$$

$$V(t, W_t) = 2\sqrt{((1+r)e^{r(T-t)} - 1)/r}\sqrt{W_t}. \quad (198)$$

Wealth Accumulation

Finally, using the optimal solution of c_t , we have

$$\frac{dW_t}{dt} = rW_t - c_t = \left(r - \frac{1}{((1+r)e^{r(T-t)} - 1)/r} \right) W_t, \quad (199)$$

$$\Rightarrow \int_0^t d \ln W_t = \int_0^t r dt + \int_0^t d \ln(1 + r - e^{-r(T-t)}), \quad (200)$$

with the initial condition $W_0 = \bar{w}$. Solving it yields:

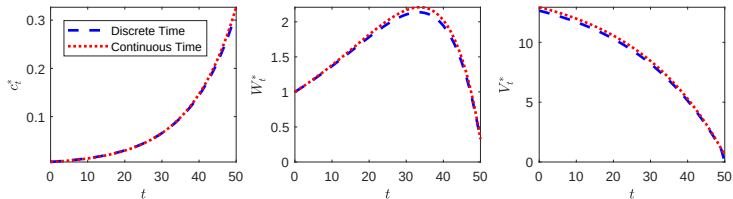
$$W_t = \frac{1 + r - e^{-r(T-t)}}{1 + r - e^{-rT}} e^{rt} \bar{w}, \quad c_t = \frac{re^{2rt}}{(1+r)e^{rT} - 1} \bar{w}. \quad (201)$$

Then,

$$\lim_{t \rightarrow T} W_t = \frac{re^{rT}}{1 + r - e^{-rT}} \bar{w}, \quad \lim_{t \rightarrow T} c_t = \frac{re^{2rT}}{(1+r)e^{rT} - 1} \bar{w}, \quad (202)$$

$$\lim_{r \rightarrow 0} W_t = \frac{T - t + 1}{T + 1} \bar{w}, \quad \lim_{r \rightarrow 0} c_t = \frac{\bar{w}}{T + 1}. \quad (203)$$

Choose $\bar{w} = 1$, $r = 0.04$, and $T = 50$.



(Here, to put things on equal footing, we plot $V/2$ instead of V under the continuous-time case.)

If you are really interested, try the following three questions.

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Question 9

Power Utility

At time $t \in [0, T]$, the investor solves:

$$V(t, W_t) = \max_{c_s, s \in [t, T]} \int_t^T \frac{c_s^{1-\gamma}}{1-\gamma} ds, \quad (206)$$

$$\text{s.t. } \frac{dW_s}{ds} = rW_s - c_s,$$

$$W_t \text{ is fixed and positive, and } W(T) \text{ free.} \quad (207)$$

Find her optimal consumption rule and value function.

A Nutshell Case

$$V(t, W_t) = \max_{c_s, s \in [t, T]} \int_t^T e^{-\rho(s-t)} \frac{c_s^{1-\gamma}}{1-\gamma} ds + \lambda e^{-\rho(T-t)} \frac{W(T)^{1-\gamma}}{1-\gamma}, \quad (208)$$

$$\begin{aligned} \text{s.t. } & \frac{dW_s}{ds} = rW_s - c_s, \\ & W_t \text{ is fixed and positive, and } W(T) \text{ free,} \end{aligned} \quad (209)$$

where $\lambda (\geq 0)$ gauges the investor's desire for leaving a bequest. The higher λ , the stronger her desire to leave a large bequest. Find her optimal consumption rule and value function.